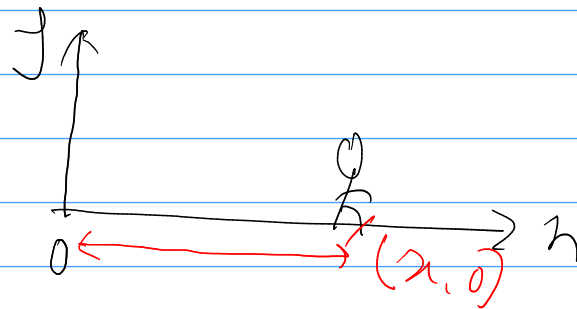
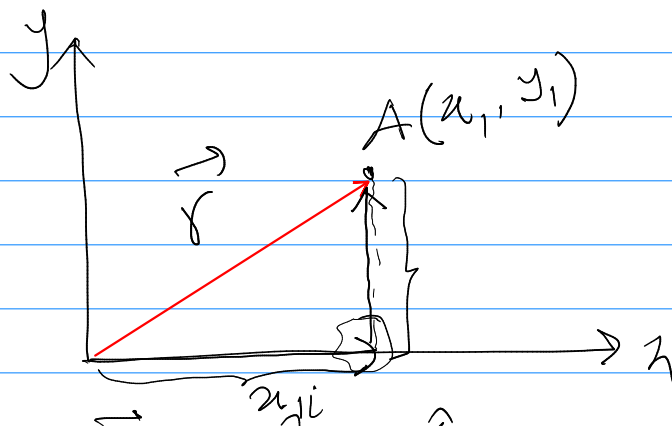


Motion in 1D and 2D



Position:-

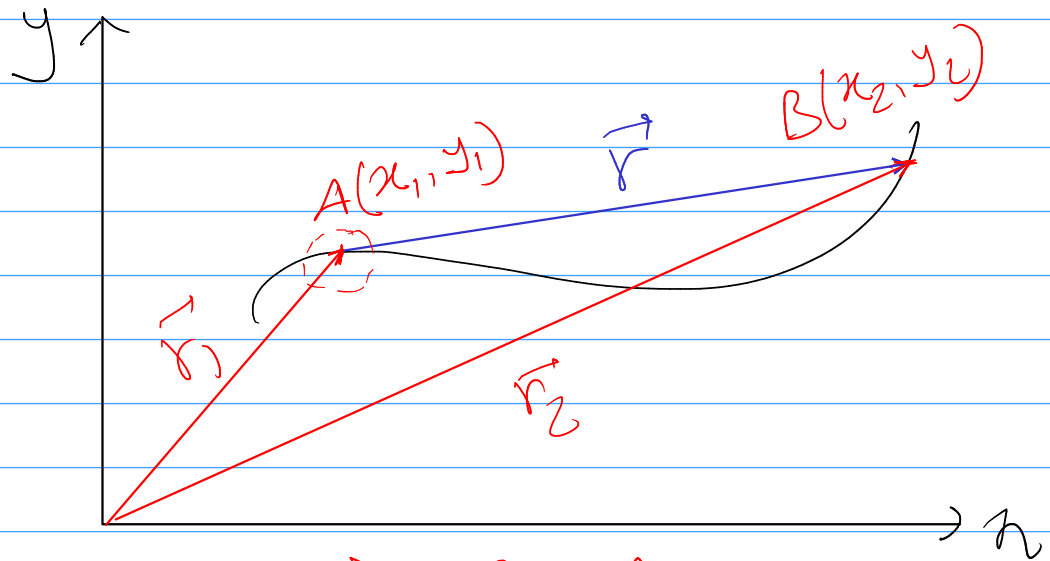
$$\vec{r} = x\hat{i} + 0\hat{j} = x\hat{i}$$



$$\vec{r} = x_1\hat{i} + y_1\hat{j}$$

$$\vec{r} = x_1\hat{i} + y_1\hat{j}$$

$$\vec{r} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$$



$$\vec{r}_1 = x_1 \hat{i} + y_1 \hat{j}$$

$$\vec{r}_2 = x_2 \hat{i} + y_2 \hat{j}$$

$$\vec{r}_1 + \vec{r} = \vec{r}_2$$

$$\Rightarrow \vec{r} = \vec{r}_2 - \vec{r}_1$$

$$\Rightarrow \vec{r} = (x_2 \hat{i} + y_2 \hat{j}) - (x_1 \hat{i} + y_1 \hat{j})$$

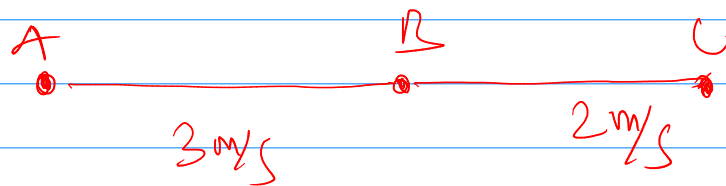
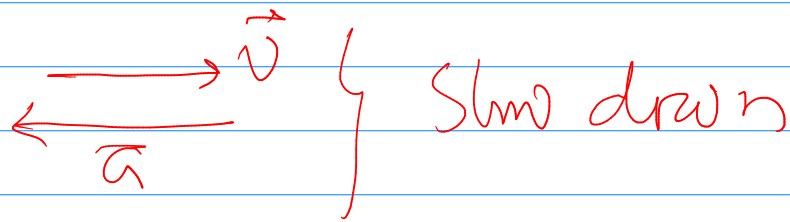
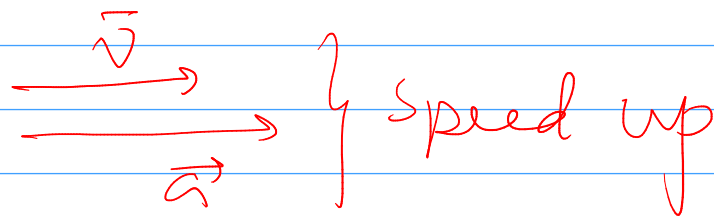
$$\Rightarrow \boxed{\vec{r} = (x_2 - x_1) \hat{i} + (y_2 - y_1) \hat{j}}$$

$$\Rightarrow \boxed{\vec{r} = \Delta x \hat{i} + \Delta y \hat{j} + \Delta z \hat{k}}$$

Velocity:

$$\vec{v} = \frac{\Delta \vec{r}}{\Delta t} \text{ m/s}$$

acceleration: $\vec{a} = \frac{\Delta \vec{v}}{\Delta t} \text{ m/s}^2$



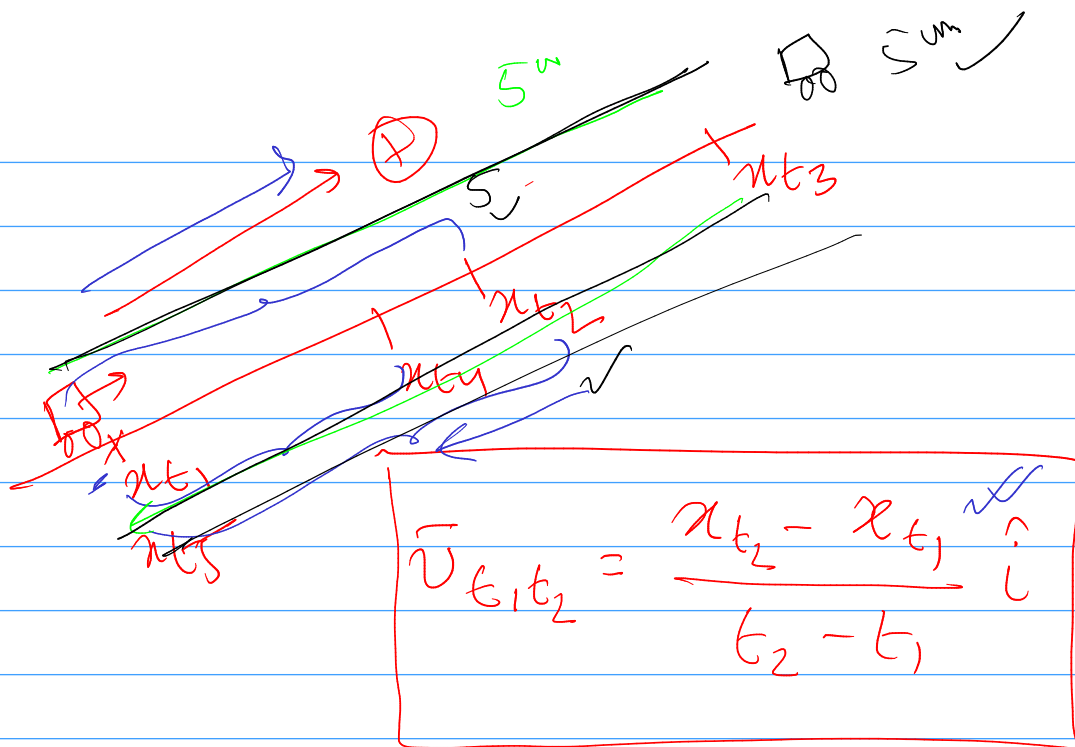
$$\bar{v} = \frac{3+2}{2} = 2.5 \text{ m/s}$$

$$\boxed{\bar{v} = \frac{\vec{v}_1 + \vec{v}_2}{2}}$$

Only if system moves at const $\vec{a}$.

Average velocity:

$$\boxed{\bar{v} = \frac{\Delta \vec{r}}{\Delta t}} \quad \checkmark$$



$$v_{t1t2} > 0$$

$$\bar{v}_{t2t4} < 0$$

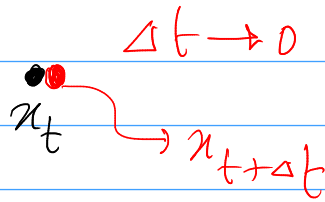
$$\bar{v}_{t1t3} = 0$$

$$\bar{v} \begin{cases} > 0 \\ = 0 \\ < 0 \end{cases}$$

Average speed:-

$$\bar{v}_{t1t2} = \frac{\text{total distance}}{\text{total time}}$$

Instantaneous Velocity:-



$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{x_{t+\Delta t} - x_t}{t+\Delta t - t} \hat{i}$$

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{x_{t+\Delta t} - x_t}{\Delta t} \hat{i}$$

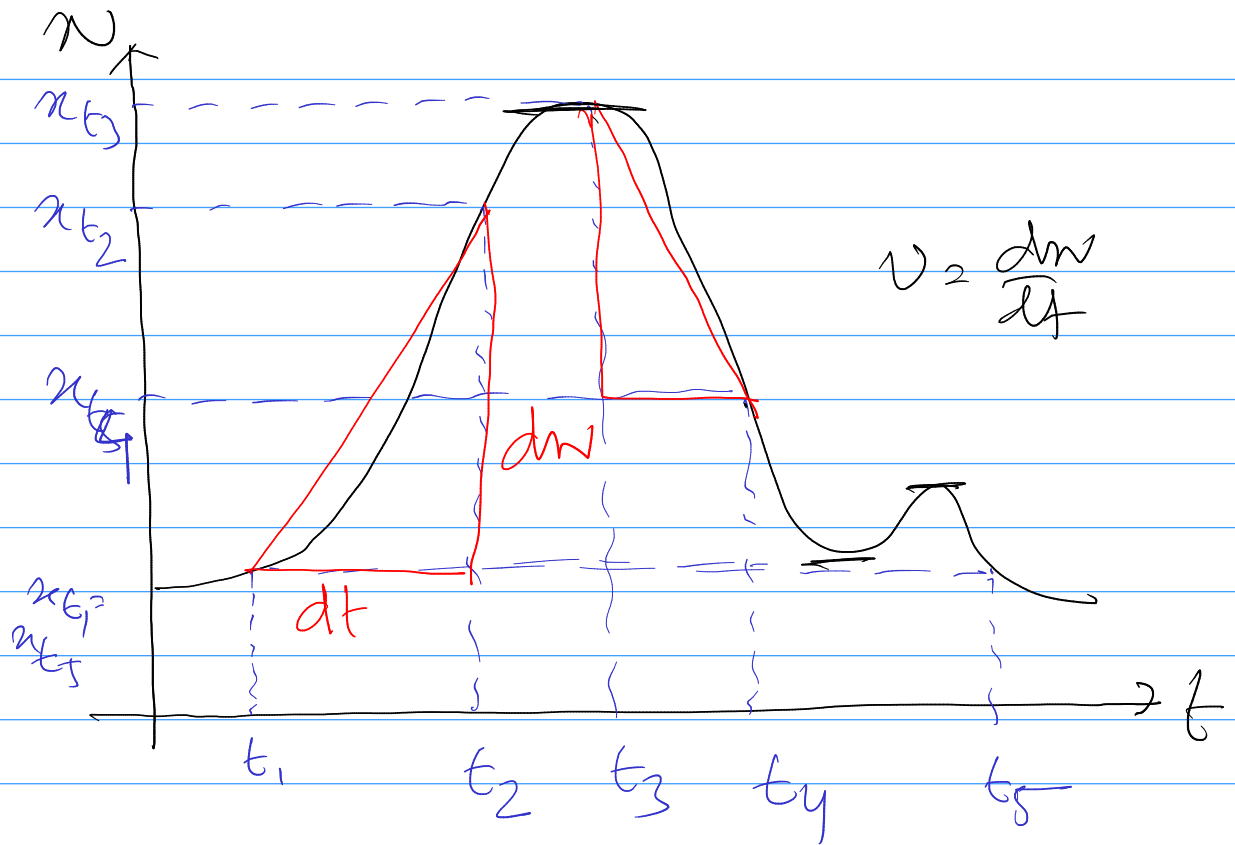
$$\vec{v} = \frac{dx}{dt} \hat{i}$$

$$\vec{v} = \frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k}$$

Average Acceleration:-

$$\vec{a}_{t_1, t_2} = \frac{\vec{v}_{t_2} - \vec{v}_{t_1}}{t_2 - t_1} = \frac{\Delta \vec{v}}{\Delta t}$$

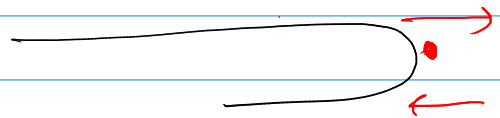
$$\vec{a} = \frac{\Delta v_x}{\Delta t} \hat{i} + \frac{\Delta v_y}{\Delta t} \hat{j} + \frac{\Delta v_z}{\Delta t} \hat{k}$$



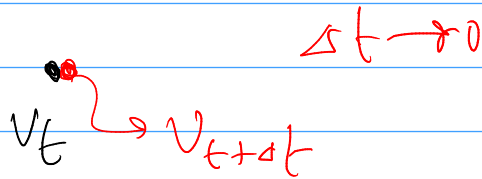
$$v_{t_1 t_2} > 0$$

$$v_{t_3 t_4} < 0$$

$$v_{t_3} = 0$$



Instantaneous acceleration:



$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\vec{v}_{t+\Delta t} - \vec{v}_t}{t + \Delta t - t}$$

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{v_{t+\Delta t} - v_t}{\Delta t} \hat{i}$$

$$\boxed{\vec{a} = \frac{dv}{dt} \hat{i}} \checkmark$$

$$\vec{v} = \frac{dv}{dt} \hat{i}$$

$$\vec{a} = \frac{d}{dt} \left(\frac{dv}{dt} \hat{i} \right) \hat{i}$$

$$\boxed{\vec{a} = \frac{d^2v}{dt^2} \hat{i}} \checkmark$$

$$\boxed{\vec{a} = \frac{dv_x}{dt} \hat{i} + \frac{dv_y}{dt} \hat{j} + \frac{dv_z}{dt} \hat{k}} \checkmark$$

$$\boxed{\vec{a} = \frac{d^2x}{dt^2} \hat{i} + \frac{d^2y}{dt^2} \hat{j} + \frac{d^2z}{dt^2} \hat{k}} \checkmark$$

$$x(t) = 8 - 6t + t^2 \text{ m} \checkmark \checkmark$$

$$v = \frac{dx}{dt} = -6 + 2t \text{ m/s} \checkmark$$

$$a = \frac{dv}{dt} = +2 \text{ m/s}^2 \checkmark \checkmark$$

at $t=0$

$$\begin{aligned} x_0 &= +8 \text{ m} \\ v_0 &= -6 \text{ m/s} \\ a_0 &= +2 \text{ m/s}^2 \end{aligned}$$

When $v=0$?

$$v = -6 + 2t = 0$$

$$\Rightarrow t = 3 \text{ s}$$

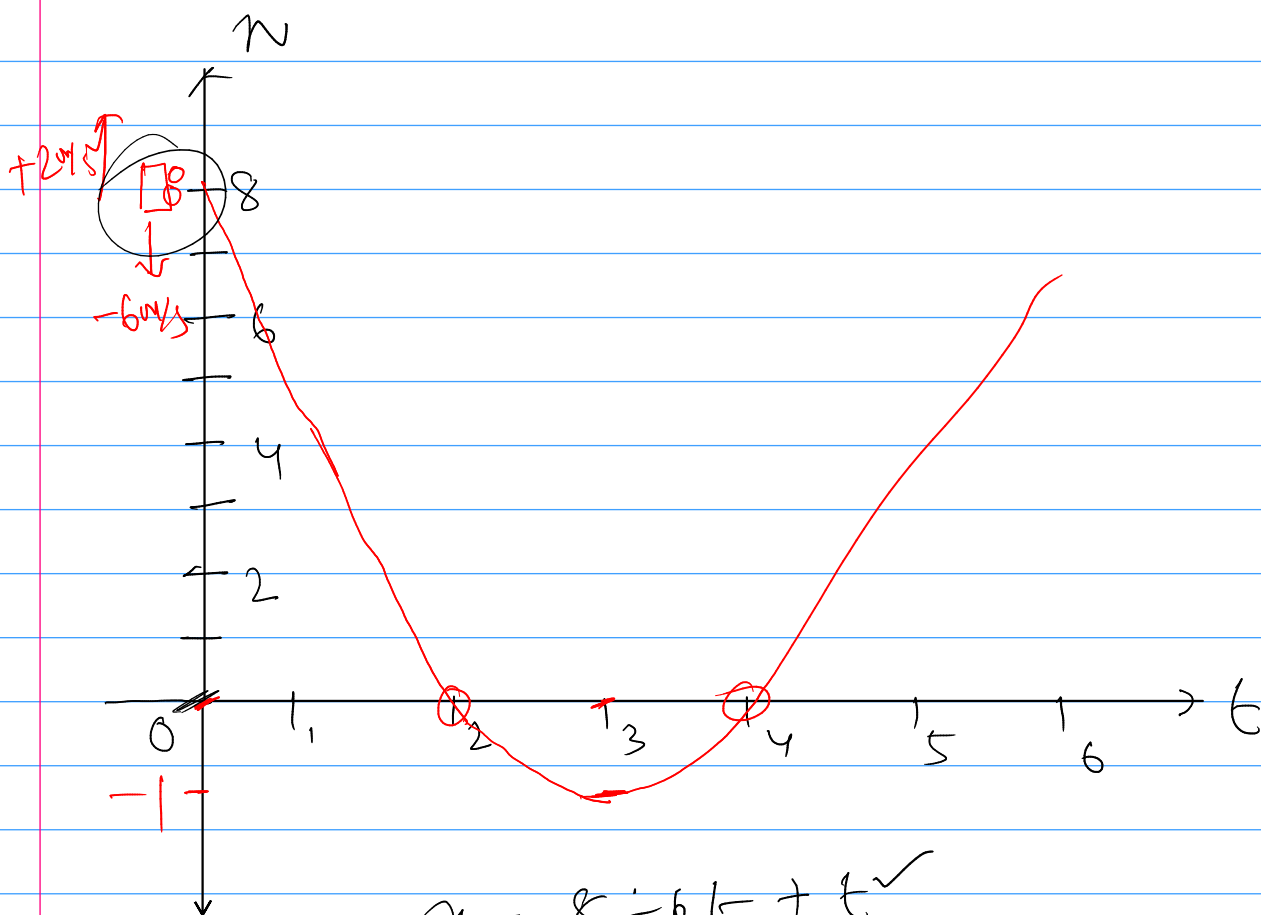
at $t = 3 \text{ s}$ what x ?

$$x = -1 \text{ m} \checkmark \checkmark$$

When $x=0$?

$$x(t) = 0 = 8 - 6t + t^2$$

$$t = +2 \text{ s}, \quad t = +4 \text{ s}$$



$$x = 8 - 6t + t^2$$

$$x = c_1 + c_2 t + c_3 t^2$$

$$v = \frac{dx}{dt} = 0 + c_2 + 2c_3 t$$

$$a = \frac{dv}{dt} = 2c_3$$

at $t = 0 \text{ s}$

$$x_0 = c_1, \quad v_0 = c_2$$

$$c_3 = \frac{1}{2} a$$

$$x = \underline{x_0} + \underline{v_0} t + \frac{1}{2} a t^2$$

$$h = v t + \frac{1}{2} a t^2$$

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \checkmark \checkmark \checkmark$$

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2 \checkmark$$

$$z = z_0 + v_{0z}t + \frac{1}{2}a_z t^2 \checkmark$$

$$\begin{cases} x = v_x t + \frac{1}{2}a_x t^2 \\ v = v_0 + a t \\ v^2 = v_0^2 + 2ax \end{cases}$$

$$v_x = \frac{dx}{dt} = 0 + v_{0x} + \frac{1}{2}a_x \cdot 2t$$

$$v_x = v_{0x} + a_x t \equiv$$

$$\boxed{x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2} \quad \begin{matrix} v = v_0 + at \\ v^2 = v_0^2 + 2ax \end{matrix}$$

$$\cancel{x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2} \quad \checkmark$$

$$x = v_{0x}t$$

$$s = vt$$

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

$$a = \frac{dv_x}{dt} = \frac{3}{2}a_0 t$$

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \checkmark$$

$$\frac{dx}{dt} = v_x$$

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \checkmark \checkmark$$

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2 \checkmark$$

$$\bar{a}_y = -9.8 \text{ m/s}^2 \hat{j}$$

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

$$\hat{j} \frac{dv_x}{dt} = 0 + \underline{v_{0x}} + \frac{3}{2}a_0 t \checkmark$$

$$\bar{a} = \frac{dv_x}{dt} = \frac{3}{2}a_0 t$$

$$\bar{a} = 3a_0 t$$

$$\bar{a} = \frac{dv}{dt} \hat{i}$$

$$\bar{v} = \frac{dr}{dt} \hat{i}$$

$$= \bar{a} = \frac{d}{dt} \left(\frac{dr}{dt} \hat{i} \right)$$

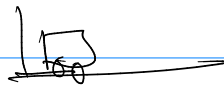
$$= \bar{a} = \frac{dr}{dt} \hat{i} + \frac{d\hat{i}}{dt}$$

$$\frac{d}{dt}(uv)$$

$$= \frac{dv}{dt} v + v \frac{d\hat{i}}{dt}$$

$$\boxed{x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2}$$

$$\boxed{h = vt + \frac{1}{2}at^2}$$



$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$$

$$z = z_0 + v_{0z}t + \frac{1}{2}a_z t^2$$

$$h = vt + \frac{1}{2}at^2$$

$$v = v_0 + at$$

$$v^2 = v_0^2 + 2ax$$

$$v_x = \frac{dx}{dt} = 0 + v_{0x} + \frac{1}{2}a_x \cdot 2t$$

$$v_x = v_{0x} + a_x t$$

$$v = v_0 + at$$

$$s = vt$$

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

$$\boxed{x = v_{0x}t}$$

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

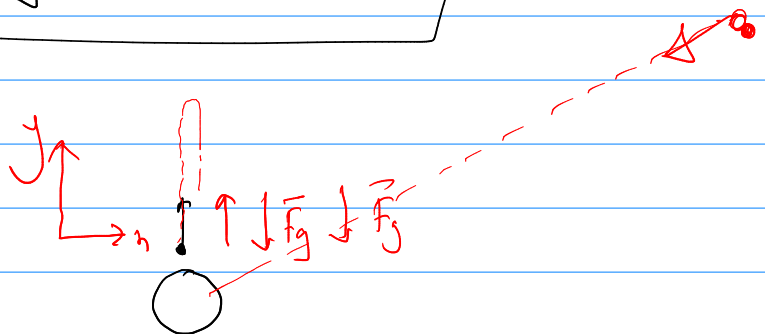
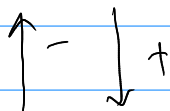
$$\frac{a_x}{1} \frac{dv_x}{dt} = \left\{ \begin{aligned} \frac{dv_x}{dt} &= 0 + \underline{v_{0x}} + \frac{1}{2}a_x 3t^2 \\ \frac{dv_x}{dt} &= 0 + 0 + \frac{3}{2}a_x t \end{aligned} \right.$$

$$a_x = 3a_x t$$

$$\boxed{x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2}$$

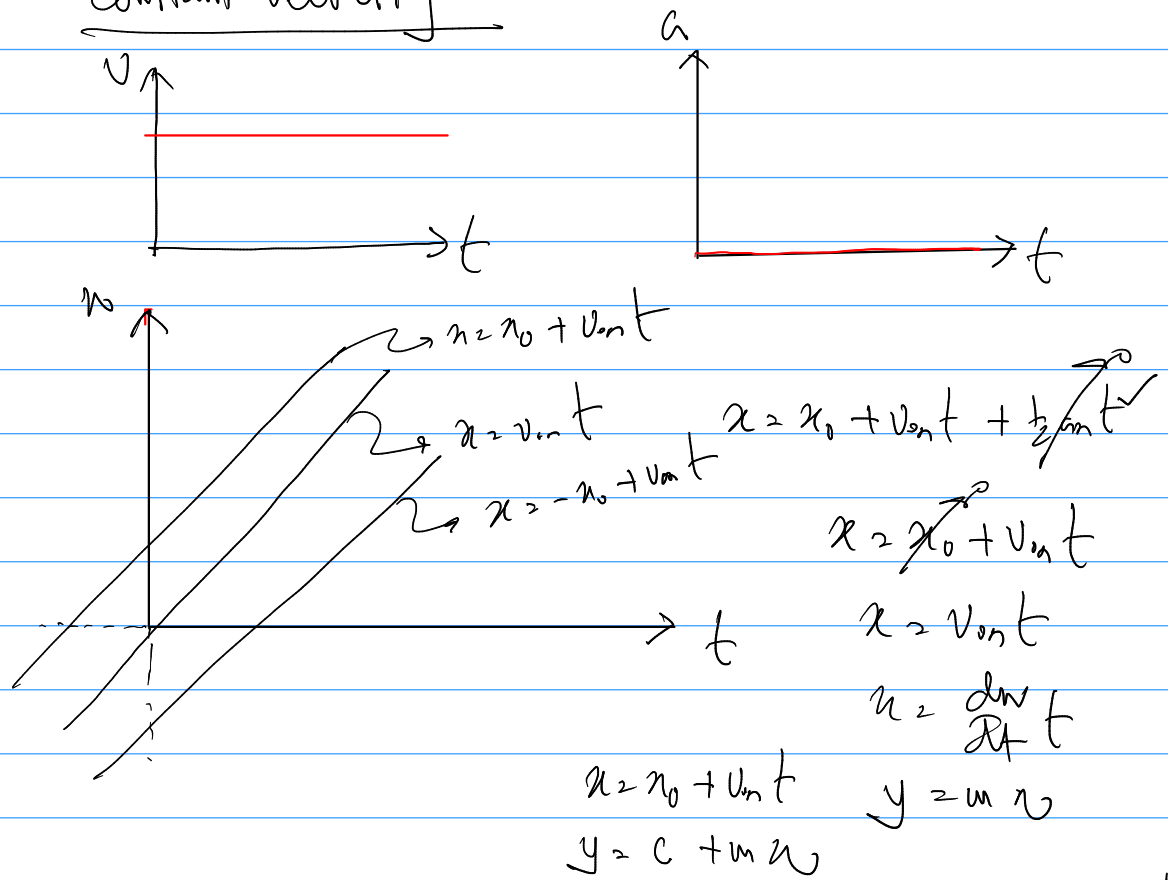
$$\boxed{y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2}$$

$$\boxed{\vec{a}_y = -9.8 \hat{j} \text{ m/s}^2}$$

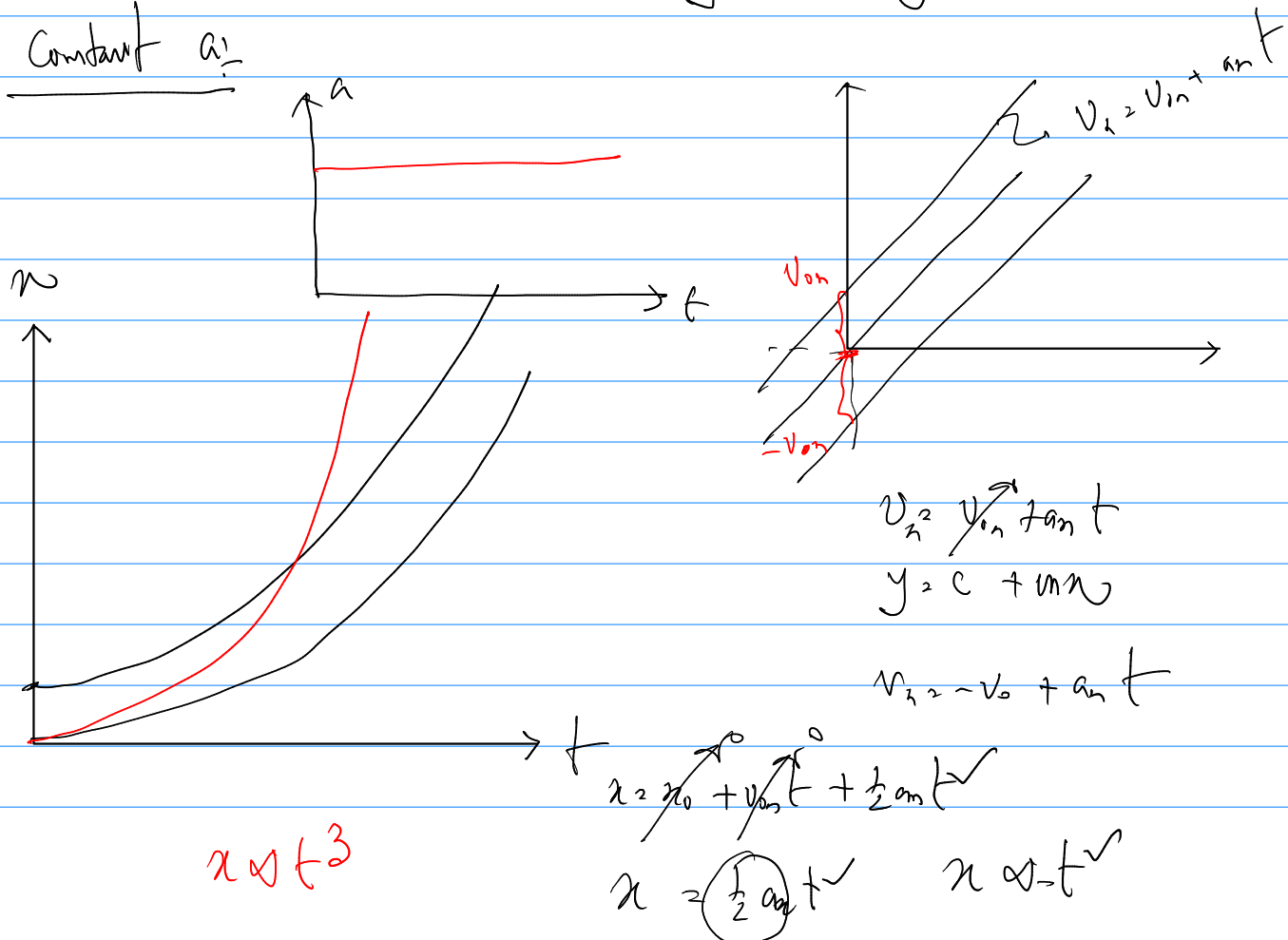


Graphical Representation:

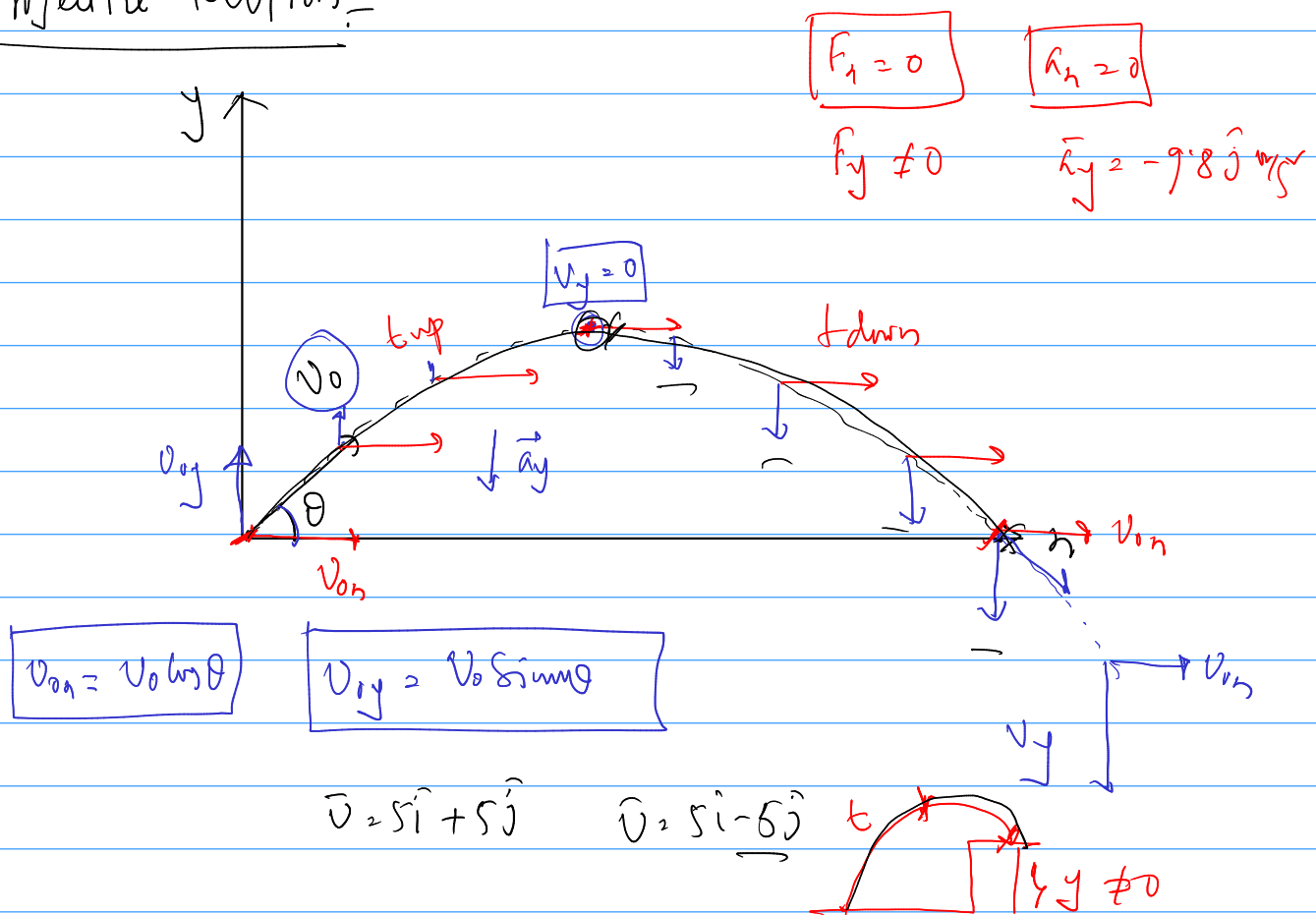
Constant velocity



Constant a_1



Projectile Motion:



(i) $t_{up} = t_{down}$ if $y = 0$

(ii) Time of flight $T_f = t_{up} + t_{down}$ if $y = 0$

$T_f = 2t_{up}$

(iii) At top $v_y = 0$

$v_x = v_{0x}$

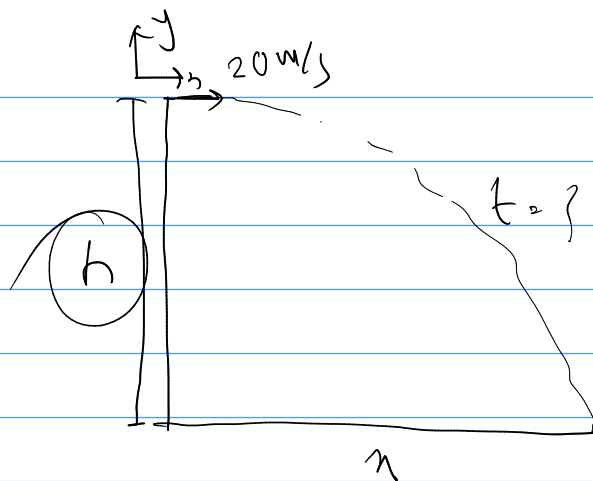
$\vec{v} = v_x \hat{i} + v_y \hat{j}$

* $t_{up} = \frac{v_0 \sin \theta}{g}$

$\vec{v} = v_x \hat{i} + v_y \hat{j}$

At top $\vec{v} = v_x \hat{i}$

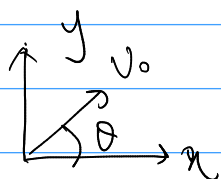
$\vec{v} = v_x \hat{i} - v_y \hat{j}$



$$T_f = \frac{2v_0 \sin \theta}{g} \quad \text{X}$$

$$T_f = 2 t_{\text{up}}$$

$$t_{\text{up}} = \frac{v_0 \sin \theta}{g}$$



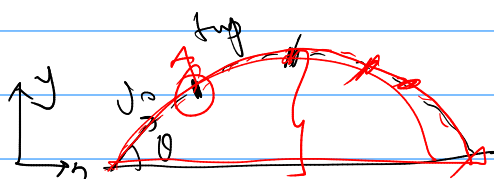
$$T_f =$$

$$T_d = T_f = \frac{v_0 \sin \theta}{g} = \frac{20 \sin \theta}{g} = 0$$



$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$$

$$\begin{cases} x = v_{0x}t \\ y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2 \end{cases}$$



$$v_y = v_0 \sin \theta - g t$$

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$$

$$0 = 0 + v_0 \sin \theta t - \frac{1}{2}g t^2$$

$$t(v_0 \sin \theta - \frac{1}{2}g) = 0$$

$$v_0 \sin \theta - \frac{1}{2}g t = 0$$

$$\begin{cases} v_y = v_{0y} + a_y t \\ v_y = v_0 \sin \theta - g t \end{cases}$$

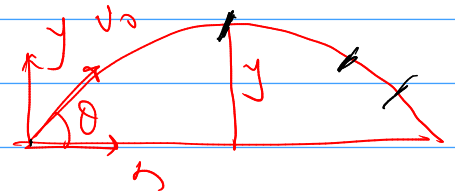
$$0 = v_0 \sin \theta - g t$$

$$t = \frac{v_0 \sin \theta}{g}$$

$$t = \frac{2v_0 \sin \theta}{g}$$

$$y = y_0 + v_{y0}t + \frac{1}{2}a_y t^2$$

$$\Rightarrow y = v_{y0}t + \frac{1}{2}a_y t^2$$



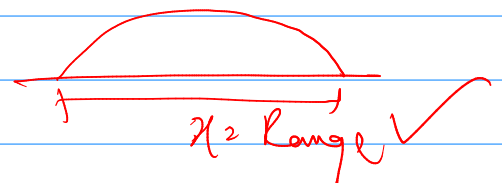
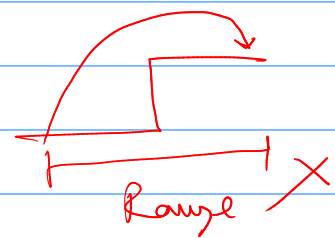
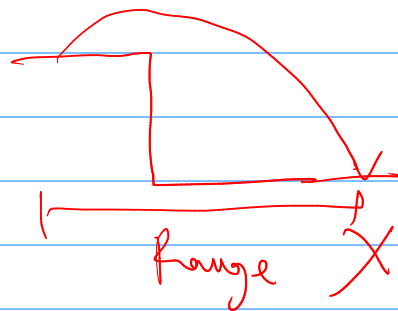
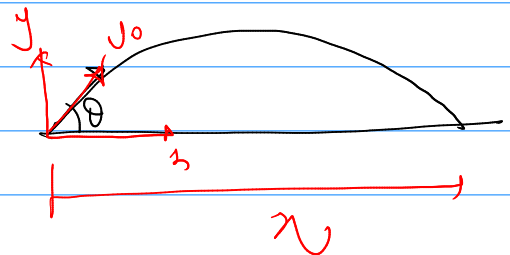
$$t_{up} = \frac{v_0 \sin \theta}{g}$$

$$\Rightarrow y = v_0 \sin \theta \times \frac{v_0 \sin \theta}{g} - \frac{1}{2}g \left(\frac{v_0 \sin \theta}{g} \right)^2$$

$$\Rightarrow y = \frac{v_0^2 \sin^2 \theta}{g} - \frac{1}{2} \frac{v_0^2 \sin^2 \theta}{g}$$

$$\Rightarrow y = \frac{v_0^2 \sin^2 \theta}{g} (1 - \frac{1}{2})$$

$$\Rightarrow y = \frac{v_0^2 \sin^2 \theta}{2g} = H$$

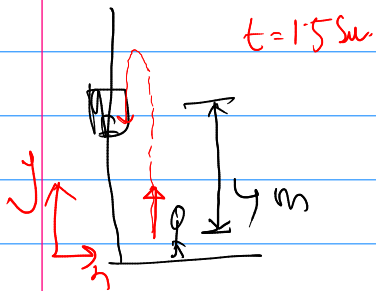
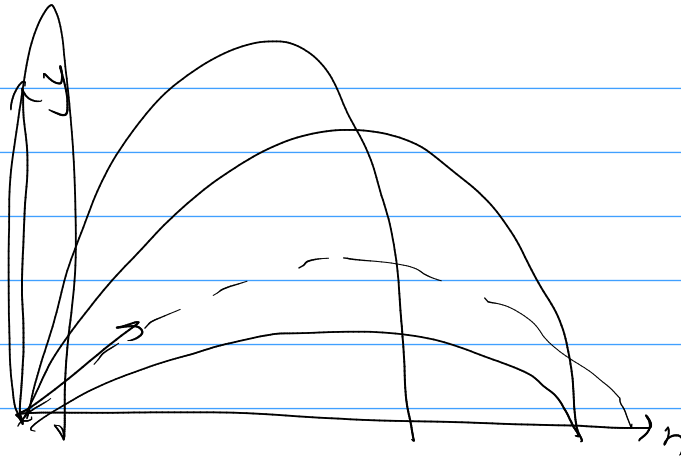


$$x = v_{x0}t$$

$$\Rightarrow x = v_0 \cos \theta \times \frac{2v_0 \sin \theta}{g}$$

$$\Rightarrow x = \frac{v_0^2 \sin 2\theta}{g}$$

$$x_{max} = \frac{v_0^2}{g}$$



$$v_y = ?$$

$$v_y = ?$$

$$v_y = v_{iy} + a_y t$$

$$y = y_0 + v_{iy} t + \frac{1}{2} a_y t^2$$

$$4 = 0 + v_{iy} \times 1.5 - 4.9 \times (1.5)^2$$

$$\Rightarrow \boxed{\vec{v}_{iy} = +10.02 \hat{j}}$$

$$v_y = v_{iy} + a_y t$$

$$\Rightarrow v_y = 10 - 9.8 \times 1.5$$

$$\Rightarrow \boxed{\vec{v}_y = -4.68 \hat{j} \text{ m/s}}$$